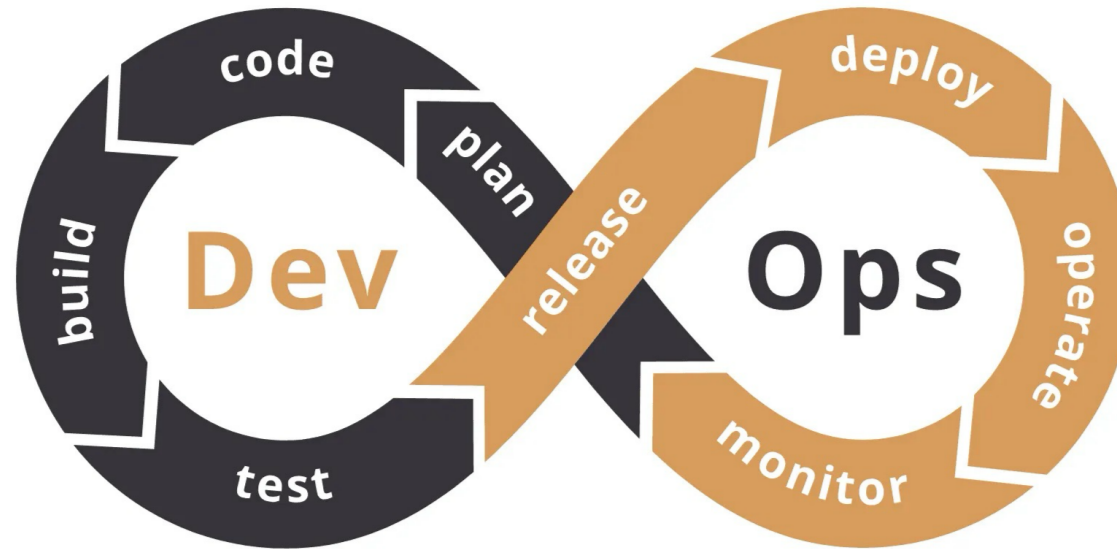




Trainer: Nilesh Ghule

Software Testing

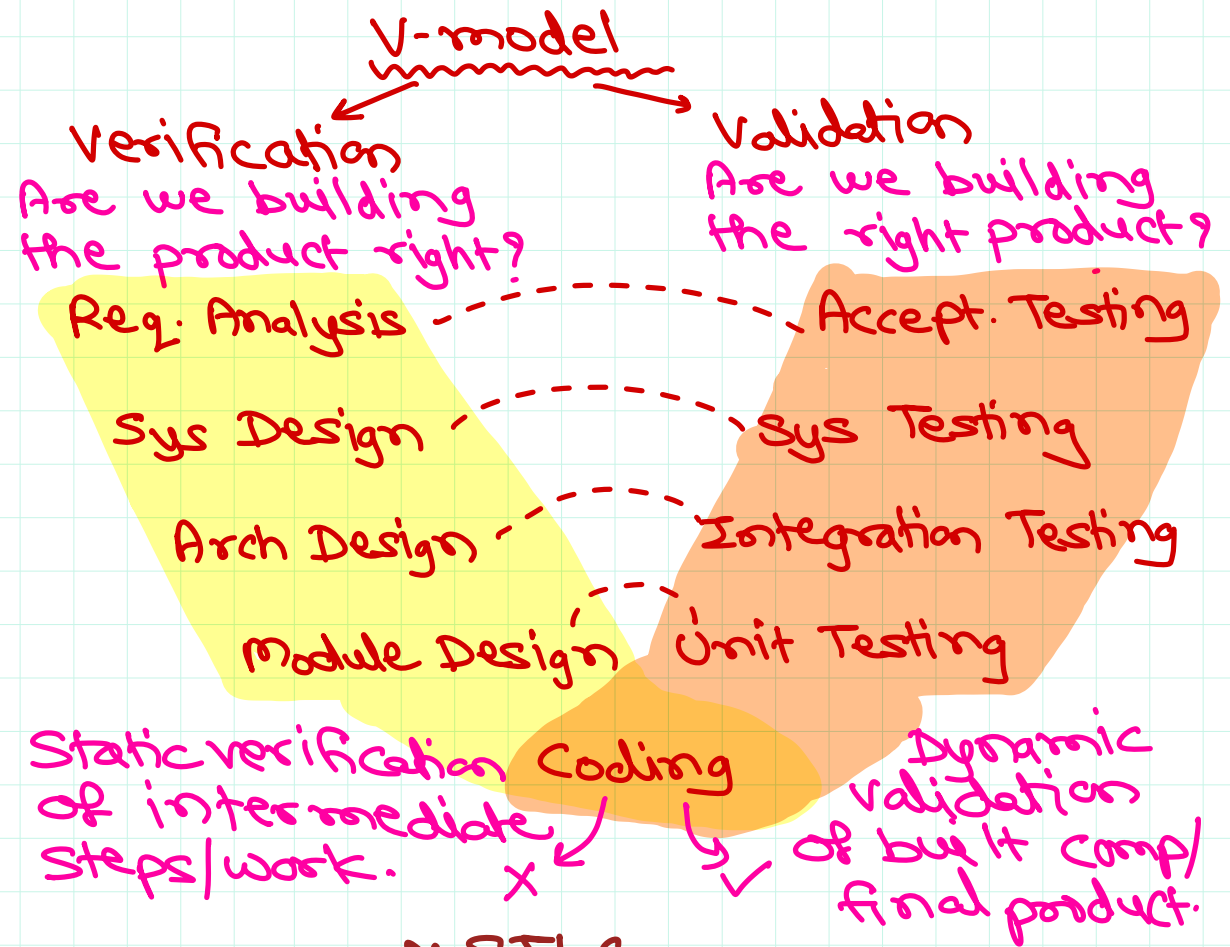
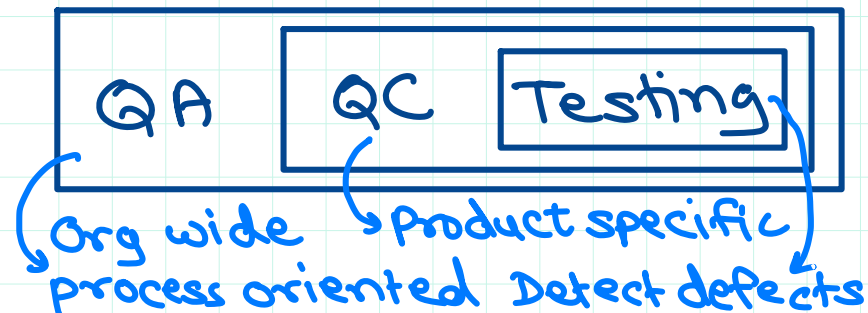


* Process of operating a system/ component under specified conditions, observing or recording the results, & making an evaluation of some aspect of the sys/component.

* Why Testing?

- ① Economic: to find bug early & fix.
- ② Product Quality & Reliability: safety critical systems.
- ③ Customer trust
- ④ Security: Testing for vulnerabilities
- ⑤ maintainability: Bug fixes & new features → Regression Testing.

* QA vs QC vs Testing



* STLC

- ① Req. analysis
↳ study SRS, init RTM
- ② Test planning
↳ scope, strategy, Res, sched
- ③ Test cases dev
↳ i/p data → expected o/p
- ④ Test Env. Setup
↳ close to prod. env.
- ⑤ Test execution
↳ Run test & record res.
- ⑥ Test Cycle Closure
↳ Summary report & Doc.

Software Testing

Manual vs Automated

- | | |
|-----------------------------|--------------------------------|
| ① Human | Scripts |
| ② Exploratory & Usability | Repetitive & Performance |
| ③ Init. cost ↓
Efforts ↑ | Tools. cost ↑
Repeat exec ↓ |
| ④ Human err | Consistent |
| ⑤ Limited cov. | More coverage |

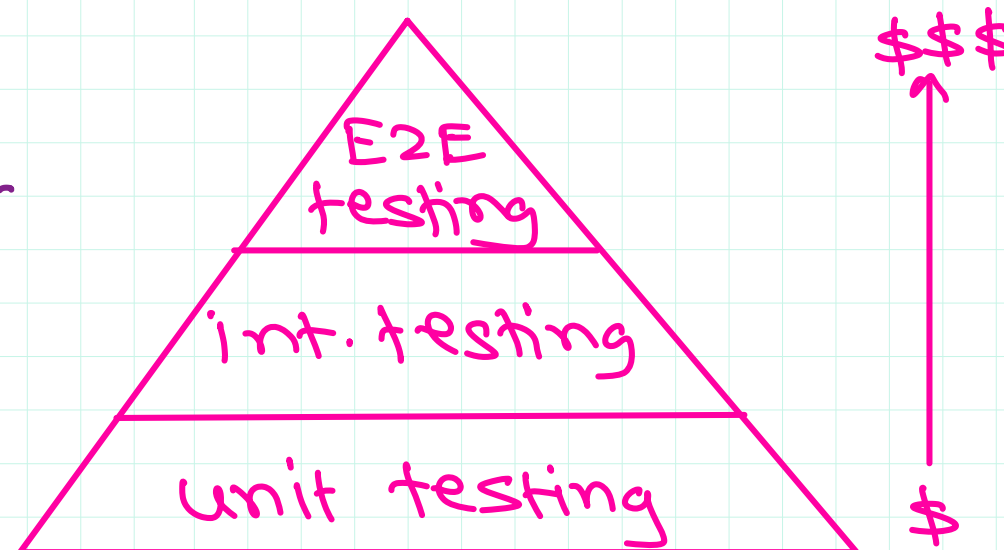
Functional Testing

- ① unit test
- ② integration test
- ③ smoke test
- ④ system test +
- ⑤ Acceptance test
 - ↳ Alpha
 - ↳ Beta
 - ↳ UAT
- ⑥ Regression test

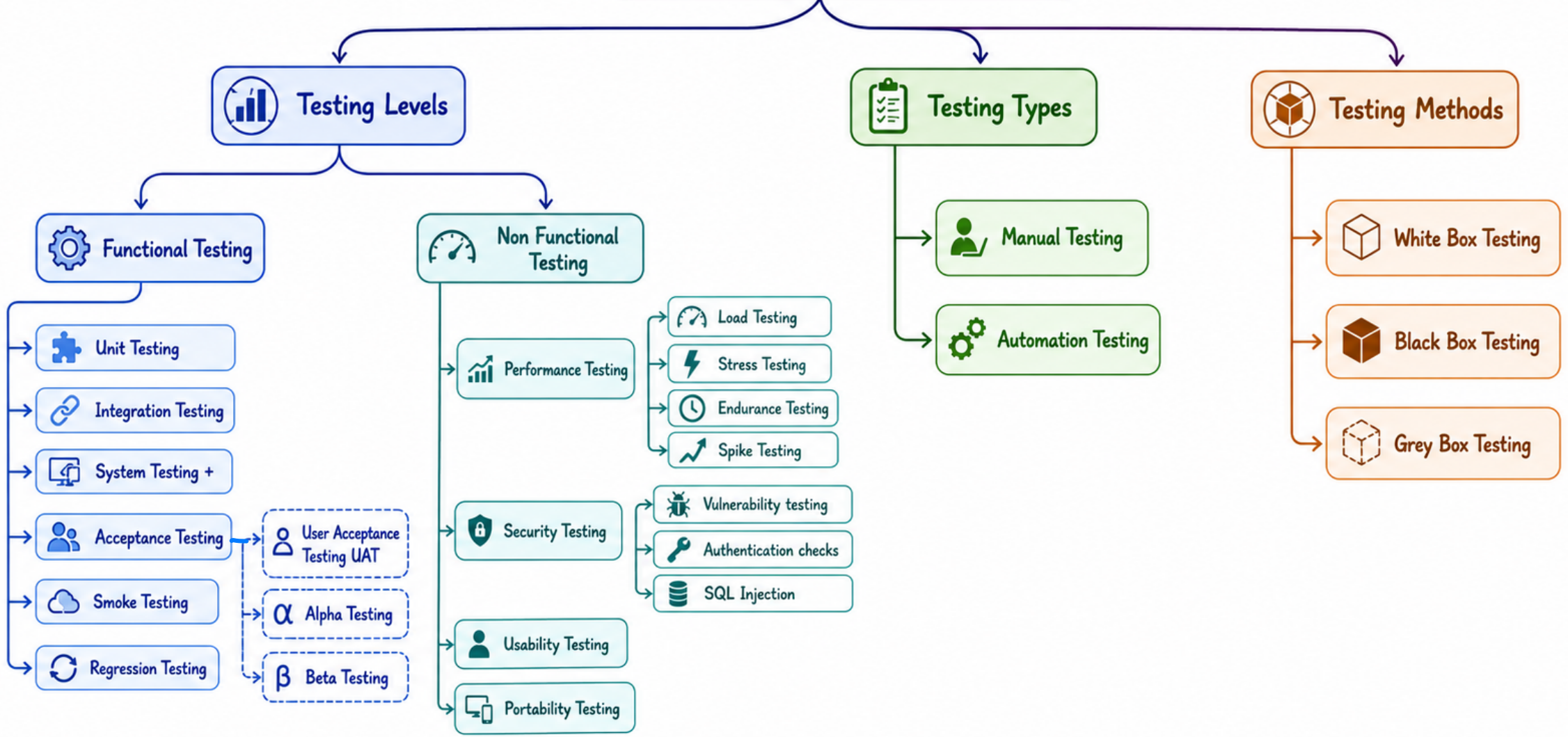
Non-Functional Testing

- ① Performance Test
 - ↳ Load test
 - ↳ Stress test
 - ↳ Soak test
 - ↳ Spike test
- ② Security Testing
- ③ Usability Testing
- ④ Portability Testing

- ### * White box vs Black Box vs Gray Box
- | | | |
|---------------|-----------------|---------------|
| ✓ Code based | ✓ No code | ✓ Tech Detail |
| ✓ by Devs | ✓ by Testers | No code |
| ✓ unit & int. | ✓ Sys & Accept. | ✓ By Testers |



Software Testing





Thank You!

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